

CODING & AI WITH PICTOBLOX



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
14 CLASSES (1 HOUR PER
CLASS)**

**MODE:
OFFLINE/ONLINE**

This course delivers a **hands-on, concept-driven learning experience** that bridges foundational coding with Artificial Intelligence through real-world applications. Each session engages learners in **building interactive programs, games, and AI-powered projects** using PictoBlox, helping them translate abstract coding concepts into meaningful digital outcomes. Students progressively explore logic, automation, sensors, and AI tools while developing problem-solving and computational thinking skills. The program uses **industry-recognized platforms and age-appropriate tools**, enabling learners to create, experiment, and innovate in a structured yet creative environment.

COURSE CURRICULUM

LEVEL 1: CODING & AI WITH PICTOBLOX

Session 1: Getting Started with PictoBlox

- Installing and setting up PictoBlox
- Exploring the interface and block categories. Understanding sprites and stage
- Creating the **Make Tobi Walk** activity

Session 2: Sprites & Animation

- Understanding sprite motion and positioning
- Working with backdrops and costumes. Creating simple animations

Session 3: Drawing with Code

- Introduction to Pen extension. Understanding coordinates and direction
- Using logic to draw shapes
- Drawing a square and triangle using code

Session 4: Variables – The Multitaskers

- Understanding variables and data storage
- Creating and updating variable values. Tracking user input using variables
- Building a sprite tracker and input-based game

Session 5: Operators in Action

- Understanding arithmetic operators
- Learning relational and logical operators
- Creating a math quiz game

Session 6: Conditional Statements

- Introduction to If–Else logic
- Creating decision-based programs
- Handling correct and incorrect conditions
- Building a **Guess the Number** game

Session 7: Loops for Automation

- Understanding repetition in programs. Using loops for automation
- Creating repeating motion and patterns
- Designing a dancing sprite or pattern animation

Session 8: Fun with Music

- Introduction to sound and music blocks
- Creating interactive sound effects
- Building a touch-based piano and dance party

Session 9: Play Time with Sensors

- Understanding sensor-based inputs
- Mapping inputs to sprite actions
- Event-based programming concepts
- Creating the **Catch the Fruit Game**

Session 10: Introduction to AI & Coding

- Understanding human vs machine intelligence
- Overview of real-world AI applications
- Connecting coding concepts to AI

- Creating an **AI Around Us** poster

Session 11: Introduction to Artificial Intelligence

- Understanding what AI is. Exploring everyday AI use cases
- Differentiating AI from traditional programs
- Activity-based AI exploration

Session 12: Introduction to Computer Vision

- Understanding image recognition concepts
- How machines identify objects and faces
- Introduction to vision-based AI models
- Identifying objects or celebrities using AI

CAPSTONE PROJECTS (ANY ONE)

Session 13 & 14:

Capstone – BMI Calculator

- Understanding problem statement and logic. Using variables, operators, and conditions
- Building a BMI calculator program
- Testing with different inputs

Capstone – Air Draw

- Understanding camera-based interaction. Using computer vision for motion tracking
- Drawing on screen using gestures
- Demonstrating final interactive AI project

LEVEL 2: CODING & AI WITH PICTOBLOX

Session 1: Face Detection

- Understanding face detection concepts
- Detecting facial features and expressions
- Mapping expressions to actions
- Creating **Face Mimic with Quarky**

Session 2: Optical Character Recognition

- Understanding text recognition using AI
- Extracting text from images
- Exploring OCR use cases
- Building **Postman – Read Text from Image**

Session 3: Speech Recognition

- Understanding speech-to-text systems
- Using voice commands in programs
- Mapping speech to actions

- Creating **Make Your Own Alexa**

Session 4: Machine Learning Basics

- Understanding training data and models
- Learning classification concepts
- Training an AI model
- Building **Cat vs Dog Classifier**

Session 5: Pose Classifier

- Understanding pose detection
- Training pose recognition models
- Testing model accuracy
- Creating **Howdy Tobi!** interaction

Session 6: Rock Paper Scissors – Part 1

- Understanding AI-based game logic
- Collecting and labeling training data
- Training the AI model
- Testing initial predictions

Session 7: Rock Paper Scissors – Part 2

- Evaluating AI model accuracy
- Improving predictions through retraining
- Handling incorrect classifications
- Refining gameplay experience

Session 8: Rock Paper Scissors – Part 3

- Integrating game logic with AI output
- Making the game fully functional
- Final testing and debugging
- Demonstrating working AI game

Session 9: Ethics in AI

- Understanding ethical concerns in AI
- Exploring bias and fairness
- Discussing responsible AI usage
- Debate on AI in education

Session 10: Real-World AI – Weather Monitoring

- Understanding environmental data collection
- Using sensors with Quarky
- Displaying real-time weather data
- Building a weather monitoring system

Session 11: Smart Waste Management

- Understanding object classification concepts
- Applying AI to waste sorting
- Designing automation logic
- Building a smart waste management system

Session 12: Home Automation System

- Understanding voice-controlled automation
- Applying NLP concepts
- Controlling devices using voice commands
- Building a smart home system

Session 13: Road Safety & Smart Cities

- Understanding AI for public safety
- Identifying road safety challenges
- Applying AI concepts to simulations
- Building a smart road safety project

Session 14: Capstone Project

- Identifying a real-world AI problem
- Selecting appropriate AI models
- Designing and implementing solution
- Presenting final AI project

Class Structure

1. Introduction to Coding & AI Concepts

- Learn the fundamentals of block-based coding, logic, and Artificial Intelligence.
- Understand how programs, data, and AI models work in real-world digital systems.

2. Guided Program Building

- Create interactive programs, animations, and games with step-by-step guidance from certified instructors.
- Apply coding concepts such as variables, conditions, loops, and events in structured activities.

3. Understanding Computational Thinking & AI Logic

- Explore how algorithms, decision-making, and automation work behind the scenes.
- Understand the basics of machine intelligence, computer vision, and sensor-based interactions.

4. Practical Coding Experiments & Challenges

- Engage in hands-on challenges that reinforce learning, including:
 - **Game Design Projects:** Apply logic, scoring, and user interaction.
 - **Automation Tasks:** Use loops and conditions to optimize programs.
 - **AI Mini-Projects:** Experiment with vision, gestures, and intelligent responses.